The Pais–Uhlenbeck Oscillator: Canonical Signs, Resonant Tori, Jordan Collision, and a Quantum Difference Spectrum

HCS-C359 source-local theorem package

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Abstract

For the two-frequency Pais–Uhlenbeck oscillator we prove an exact Ostrogradsky normal form and classify every classical orbit closure. A rational frequency ratio closes every orbit; an irrational ratio leaves only the equilibrium and single-mode circles periodic, while each genuine double-mode orbit is dense on a two-torus. We then resolve the equal-frequency Jordan collision and all real sign degenerations. Finite exact receipts audit conventions; the continuum statements are proved analytically.

1 Frozen model and canonical sign

Consider

$$L = \frac{1}{2}\{\ddot{x}^2 - (\omega_1^2 + \omega_2^2)\dot{x}^2 + \omega_1^2\omega_2^2 x^2\}, \qquad 0 < \omega_1 < \omega_2. \tag{1}$$

The higher-order Euler–Lagrange equation factors as

$$(D^2 + \omega_1^2)(D^2 + \omega_2^2)x = 0. \tag{2}$$

With $q_0 = x, q_1 = \dot{x}, p_1 = \ddot{x}$ and $p_0 = -(\omega_1^2 + \omega_2^2)\dot{x} - x^{(3)}$, the Ostrogradsky Hamiltonian is

$$H = p_0 q_1 + \frac{1}{2}p_1^2 + \frac{1}{2}(\omega_1^2 + \omega_2^2)q_1^2 - \frac{1}{2}\omega_1^2\omega_2^2 q_0^2. \tag{3}$$

Put $\Delta = \omega_2^2 - \omega_1^2$ and define

$$\begin{aligned} Q_1 &= (p_1 + \omega_2^2 q_0)/\sqrt{\Delta}, & P_1 &= (p_0 + \omega_1^2 q_1)/\sqrt{\Delta}, \\ Q_2 &= (p_1 + \omega_1^2 q_0)/\sqrt{\Delta}, & P_2 &= -(p_0 + \omega_2^2 q_1)/\sqrt{\Delta}. \end{aligned} \tag{4}$$

Direct Poisson-bracket calculation gives $\{Q_j, P_k\} = \delta_{jk}$ and zero cross brackets. Substitution gives

$$H = -\frac{1}{2}(P_1^2 + \omega_1^2 Q_1^2) + \frac{1}{2}(P_2^2 + \omega_2^2 Q_2^2). \tag{5}$$

Thus our convention has a negative low-frequency mode and a positive high-frequency mode. This explicit check is the *canonical-sign owner*; changing either sign changes the theorem.

Theorem 1 (Complete classical closure)**.** *In the chamber above, the linear Hamiltonian flow is complete. If $\omega_1/\omega_2 \in \mathbb{Q}$, every trajectory is periodic. If the ratio is irrational, exactly the equilibrium and single-mode trajectories are periodic, and each double-mode trajectory is dense on its invariant two-torus.*

Proof. Equation (2) has the general solution

$$x = a_1 \cos\omega_1 t + b_1 \sin\omega_1 t + a_2 \cos\omega_2 t + b_2 \sin\omega_2 t. \tag{6}$$

Equivalently, (5) preserves the two modal radii $R_j^2 = Q_j^2 + P_j^2/\omega_j^2$. If $\omega_1 = gm, \omega_2 = gn$ with coprime positive m, n, then $T = 2\pi/g$ advances the two phases by $2\pi m, 2\pi n$. It is a common period; a collapsed one-mode circle can have a smaller least period.

For irrational ratio, a periodic double-mode point would require both $\omega_j T \in 2\pi\mathbb{Z}$, contradicting irrationality. Single-mode circles retain period $2\pi/\omega_j$. Kronecker's theorem makes the two-frequency phase orbit dense in $S^1 \times S^1$, and the invertible map (4) carries that closure to the Ostrogradsky torus. Constant coefficients give completeness for all real time. QED

Source and claim boundary

Pais and Uhlenbeck introduced the non-local-action model in *Physical Review* 79 (1950), 145–165, doi:10.1103/PhysRev.79.145. Smilga analyzed higher-derivative ghost dynamics in *Nuclear Physics B* 706 (2005), 598–614, doi:10.1016/j.nuclphysb.2004.10.037. Bolonek and Kosiński studied Hamiltonian structures in arXiv:quant-ph/0501024. These sources establish lineage; every displayed claim above is rederived for the frozen convention. No priority claim is made.